Experiment 6

Full Adder using different Modelling Styles

Aim: To design and simulate a Full Adder using Hardware Design Language

(HDL)

Apparatus: Vivado Tool.

Theory:

A **Full Adder** is a fundamental combinational circuit used in digital electronics for Arithmetic Operations. A full adder performs the addition of three binary bits.

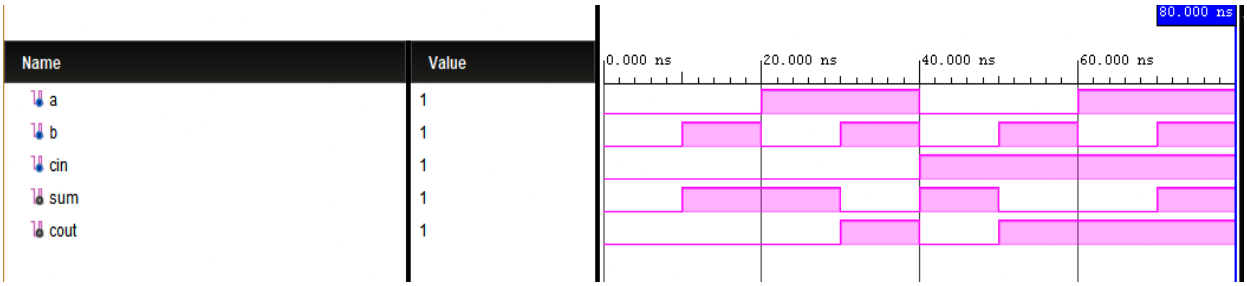
Full Adder: A full adder has three inputs: two significant bits (A and B) and a carry-in (Cin) from the previous stage. It produces two outputs: a sum (S) and a carry-out (Cout).

(i) Full Adder using Data Flow Modelling

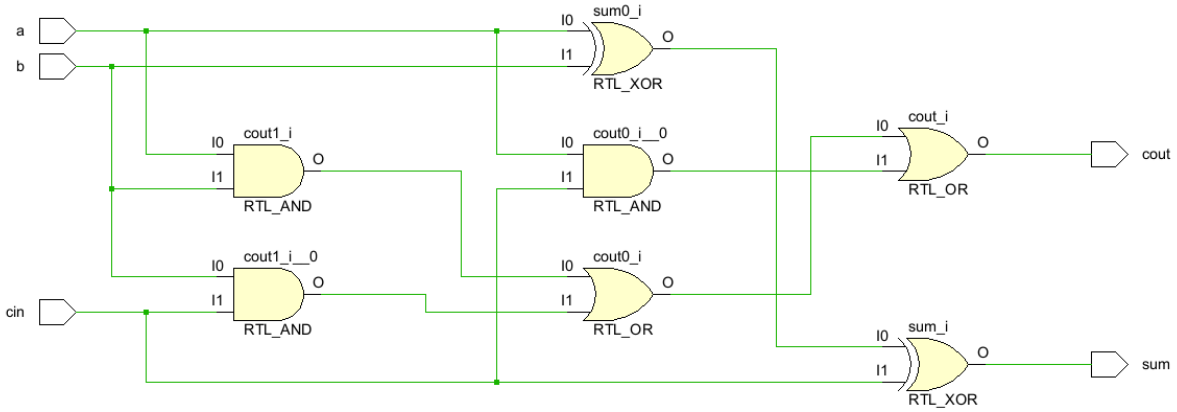
Verilog Code:

```
1
2  module full_adder(
3      input a,
4      input b,
5      input cin,
6      output cout,
7      output sum
8      );
9      assign sum = a^b^cin;
10     assign cout = (a & b) | (b & cin) | (a & cin);
11 endmodule
12
```

Simulation:



Schematic:

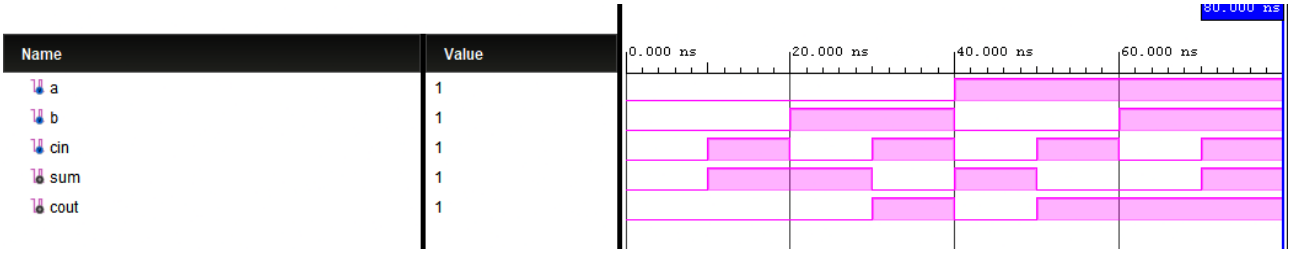


(ii) Full Adder using Behavioral Modelling

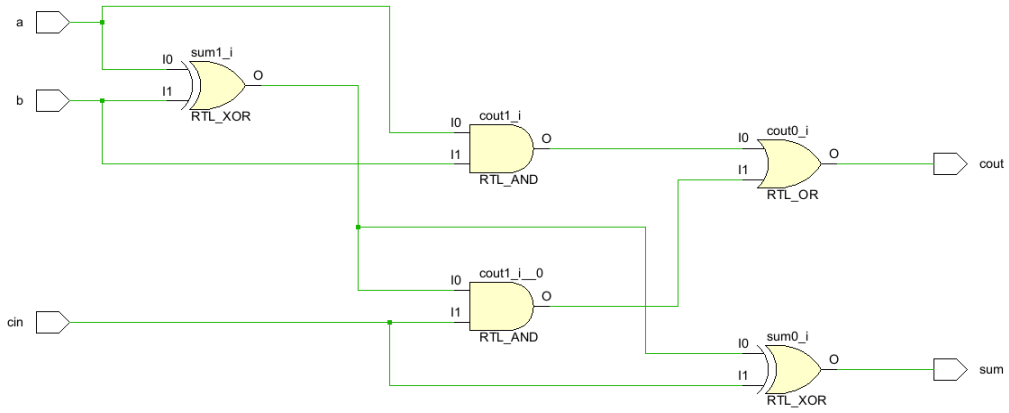
Verilog Code:

```
1
2   module full_adder( a, b, cin, sum, cout );
3       input wire a, b, cin;
4       output reg sum, cout;
5
6       always @(a or b or cin)
7           begin
8               sum = a ^ b ^ cin;
9               cout = a&b | (a^b) & cin;
10      end
11  endmodule
12
```

Simulation:



Schematic:



(iii) Full Adder using Structural Modelling

Verilog Code:

```
module FA_Str(
    input A, B, Cin,
    output Sum, Cout
);
    wire w1, w2, w3;

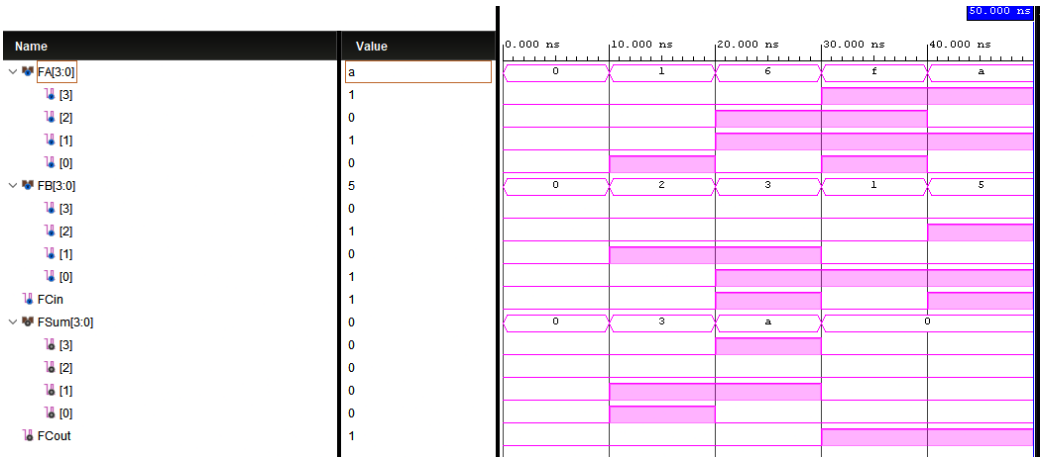
    xor (w1, A, B);
    xor (Sum, w1, Cin);
    and (w2, w1, Cin);
    and (w3, A, B);
    or  (Cout, w2, w3);

endmodule

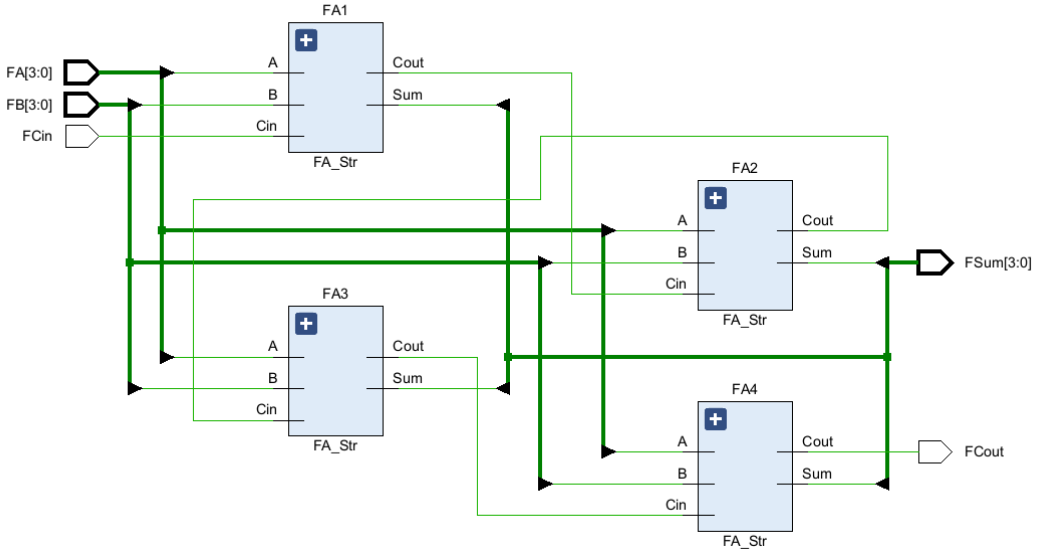
module full_adder(
    input [SIZE-1:0] FA, FB,
    input FCin,
    output [SIZE-1:0] FSum,
    output FCout
);

    parameter SIZE = 4;
    wire [SIZE-2:0] FTemp;
    FA_Str FA1 (.A(FA[0]), .B(FB[0]), .Cin(FCin),    .Sum(FSum[0]), .Cout(FTemp[0]));
    FA_Str FA2 (.A(FA[1]), .B(FB[1]), .Cin(FTemp[0]), .Sum(FSum[1]), .Cout(FTemp[1]));
    FA_Str FA3 (.A(FA[2]), .B(FB[2]), .Cin(FTemp[1]), .Sum(FSum[2]), .Cout(FTemp[2]));
    FA_Str FA4 (.A(FA[3]), .B(FB[3]), .Cin(FTemp[2]), .Sum(FSum[3]), .Cout(FCout));
endmodule
```

Simulation:



Schematic:



Procedure

1. Open **Vivado Design Suite** and create a new RTL project by specifying the project name, location, and required FPGA device.
2. Create a new **Verilog source file** and define the three inputs **A, B, Cin** and two outputs **Sum, Cout** for the full adder.
3. Implement the full adder using **Data Flow Modelling** by writing continuous assign statements for the Sum and Carry outputs.
4. Create a **testbench** for the data flow model and apply the required combinations of A, B, and Cin inputs.
5. Run **Behavioral Simulation** in Vivado and observe the simulation waveform to verify the Sum and Carry outputs.
6. Implement the full adder using **Behavioral Modelling** using an always block, with Sum and Cout assigned according to the full-adder logic.
7. Simulate the behavioral model and verify its waveform and RTL schematic with the expected full-adder operation.
8. Implement the full adder using **Structural Modelling** by connecting basic logic gates such as XOR, AND, and OR to form the full-adder circuit.
9. Simulate the structural model and observe the waveform and schematic to verify the correct propagation of Sum and Carry.
10. Compare the simulation results of all three modelling styles and verify that they produce the expected full-adder outputs for the applied input combinations.

Conclusion

The full adder was successfully designed and simulated in Vivado using **Data Flow, Behavioral, and Structural modelling styles**. The simulation waveforms and RTL schematics verified the correct operation of the full adder, producing the expected **Sum and Carry-out** for the applied combinations of A, B, and Cin.